Ananya Bandopadhyay

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ananyabandopadhyay.github.io

EDUCATION

Syracuse University | *Ph.D. in Physics* 2021 – 2027 (expected)

- NASA FINESST fellow
- Research Area: High Energy Astrophysics
- Advisor: Dr. Eric Coughlin

Indian Institute of Science, Education & Research, Pune, India | 2016–2021
B.S.-M.S. Dual Degree in Physics

- Research Area: Gravitational Wave Astronomy
- M.S. Thesis: Parameter Estimation for Binary Black Hole Populations using GW observations from Individual Mergers and the Stochastic Background
- Advisor: Dr. Sanjit Mitra

RESEARCH INTERESTS

High-Energy Astrophysical Phenomena

- Tidal Disruption Events
- Compact Binary Mergers.
- Tidal Interactions in Compact Binary Systems.

PUBLICATIONS AS LEAD/CORRESPONDING AUTHOR

1. **Bandopadhyay, A.**, Amend, B., Coughlin, E. R., Nixon, C. J., Pasham, D. R,. Wevers, T., The Role of Stellar Spin in Repeating Partial Tidal Disruption Events, ApJ, 1007, 181, 2026.
   Data-release repository: zenodo.org/records/20594641
2. **Bandopadhyay, A.**, Coughlin, E. R., Nixon, C. J., The Maximum Gravity Model for partial Tidal Disruption Events: Mass Loss, Peak Fallback Rate and Dependence on Stellar Properties, ApJ, 998, 81, 2026.
   Data-release repository: zenodo.org/records/17822028
3. Fancher, J.†, **Bandopadhyay, A.**, Coughlin, E. R., Nixon, C. J., Predicting the Properties of the Fallback Rate from Tidal Disruption Events: Investigating the Maximum Gravity Model, ApJ, 990, 104, 2025.
4. **Bandopadhyay, A.**, Coughlin, E. R., Nixon, C. J., Repeated Tidal Interactions between Stars and Supermassive Black Holes: Mass Transfer, Stability and Implications for Repeating Partial Tidal Disruption Events, ApJ, 987, 16, 2025.
5. **Bandopadhyay, A.**, Coughlin, E. R., Nixon, C. J., Pasham, D. R., Repeating nuclear transients from repeating partial tidal disruption events: reproducing ASASSN-14ko and AT2020vdq, ApJ, 974, 80, 2024.

† = mentored student

6. **Bandopadhyay, A.**, Kacanja, K., Somasundaram R., et al., Measuring Neutron Star Radius with second and third generation Gravitational Wave Detector Networks, Class. Quantum Grav, 41, 225003, 2024.
   Data-release repository: github.com/sugwg/bns-eos-nggw
7. **Bandopadhyay, A.**, Fancher, J.†, Athian, A.†, Indelicato, V.†, Kapalanga, S.†, Kumah, A.†, et al., The Peak of the Fallback Rate from Tidal Disruption Events: Dependence on Stellar Type, ApJ, 961, L2, 2024.
8. **Bandopadhyay, A.**, Reed, B., Padamata, S., Leon, E., et al., Detectability of sub-solar mass neutron stars through a template bank search, Phys. Rev. D, 107(10):103012, 2023.
   Data-release repository: github.com/sugwg/sub-solar-ns-detectability

PUBLICATIONS AS CONTRIBUTING AUTHOR

1. O'Connor, B., Gill, R., DeLaunay, J., Hare, J., Pasham, D., Coughlin, E. R., **Bandopadhyay, A.**, et al., Comprehensive X-Ray Observations of the Exceptional Ultralong X-Ray and Gamma-Ray Transient GRB 250702B with Swift, NuSTAR, and Chandra: Insights from the X-Ray Afterglow Properties, ApJ, 994, L17, 2025.
2. Makrygianni, L., Arcavi, I., Newsome, M., **Bandopadhyay, A.**, et al., The Double Tidal Disruption Event AT 2022dbl Implies that at Least Some "Standard" Optical Tidal Disruption Events Are Partial Disruptions, ApJ, 987, L20, 2025.
3. Gupta, I., Afle, C., Arun, K.G., **Bandopadhyay, A.**, et al., Characterizing Gravitational Wave Detector Networks: From A$^\sharp$ to Cosmic Explorer, Class. Quantum Grav. 41, 245001, 2024.
4. Pasham, D. R., Coughlin, E. R., Guolo, M., et. al, (incl) **Bandopadhyay, A.**, A Potential Second Shutoff from AT2018fyk: An Updated Orbital Ephemeris of the Surviving Star under the Repeating Partial Tidal Disruption Event Paradigm, ApJ, 971, L31, 2024.

CONFERENCE PROCEEDINGS

1. **Bandopadhyay, A.**, Coughlin, E. R., Fancher, J., et. al., Repeating nuclear transients from repeating partial tidal disruption events, Astronomische Nachrichten, 347, 5, e70083, 2026.

BOOK CHAPTERS

1. **Bandopadhyay, A.**, Liu C., Pan, Z., Yao, P. & Shen, R., *Theoretical Models for Repeating Partial TDEs*, invited review chapter for the book "Mysterious Repeating Signals from Centers of Galaxies" (Eds. Dheeraj Pasham, Eric Coughlin, Riccardo Arcodia, Petra Sukova, Itai Linial, Springer Singapore, expected in 2027).

MANUSCRIPTS IN PREP

1. **Bandopadhyay, A.**, Indelicato, V.†, Coughlin, E. R., Nixon, C.J., *Flares from the Less Bound Debris in Repeating Partial Tidal Disruption Events*, to be submitted to ApJ (December 2026).
2. **Bandopadhyay, A.**, & Coughlin, E. R., *A Generalized Equilibrium Model for Rotating Stars*, to be submitted to ApJ (December 2026).

SCIENTIFIC PRESENTATIONS

- Talk: "Repeating Nuclear Flares from TDEs – Constraints on Stellar Properties"
  Columbia THEA group meeting. November 13, 2026
- Talk: "Repeating Nuclear Flares from TDEs – Constraints on Stellar Properties"
  Flatiron Stars and Plasma Astrophysics group meeting. November 12, 2026
- Talk: "Repeating Nuclear Flares from TDEs – Constraints on Stellar Properties"
  UVA/NRAO Lunch Talk. Oct 23, 2026
- Talk: "Repeating Nuclear Flares from TDEs – Constraints on Stellar Properties"
  UW Madison Astrophysics seminar. Oct 5, 2026
- Talk: "Repeating Nuclear Flares from TDEs – Constraints on Stellar Properties"
  TDEs and NTs in Crete 2026, Crete. August 26, 2026
- Talk: "Repeating Nuclear Flares from TDEs"
  Cornell & Syracuse joint group meeting. August 4, 2026
- Poster: "Repeated Tidal Interactions between Stars and Supermassive Black Holes: Mass Transfer, Stability and Implications for Repeating Partial Tidal Disruption Events"
  AAS HEAD 22nd Meeting, St. Louis, MO. October 2025
- Talk: "Repeating Nuclear Transients from Repeating Partial Tidal Disruption Events"
  Astrophysics Group Meeting, Princeton University. September 3, 2025
- Talk: "Repeating Nuclear Transients from Repeating Partial Tidal Disruption Events"
  Cornell & Syracuse joint group meeting. August 11, 2025
- Talk: "Repeating Nuclear Transients from Repeating Partial Tidal Disruption Events"
  X-ray QPEs and Repeating Nuclear Transients, Madrid. June 16, 2025
- Talk: "Repeating Nuclear Transients from Repeating Partial Tidal Disruption Events"
  AAS 245th Meeting, National Harbor, MD. January 16, 2025
- Talk: "The Peak of the Fallback Rate from Tidal Disruption Events: Dependence on Stellar Type"
  KITP Conference: "Anticipating the Rising Tide of Tidal Disruption Events: Theory and Observations", link. April 25, 2024
- Talk: "Close Encounters of the Supermassive Black Hole Kind: Tidal Disruption Events and What They Can Reveal About Black Holes and Stars in Distant Galaxies"
  AAS 243 Press Conference: Oddities in the Sky, link. January 11, 2024
- iPoster: "Peak Fallback Rate from Tidal Disruption Events: Dependence on Stellar Type"
  AAS 243rd Meeting, New Orleans, LA. January 9, 2024
- Paper Talk: "The Peak of the Fallback Rate from Tidal Disruption Events: Dependence on Stellar Type"
  AAS Journal Author Series, YouTube. January 2024
- Talk: "Using HTCondor to Study Gravitational Waves from Binary Neutron Star Mergers"
  OSG User School, Madison, WI July 2022

AWARDS & FELLOWSHIPS

- Syracuse University Research Excellence Dissertation Grant. 2026
  - Awarded to 10 students across all departments | $3,000.00
- NASA FINESST fellowship 2024–2027
  - FI: A. Bandopadhyay, PI: E. R. Coughlin | $147,986.00
- Henry Levinstein fellowship, Department of Physics, Syracuse University. 2021
- Indian Academy of Sciences' Summer Research Fellowship 2018

- Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship 2017–2021
  - *Awarded by the Department of Science & Technology, Government of India*
- INSPIRE Scholarship for Higher Education (SHE) Fellowship 2016
  - *Awarded by the Department of Science & Technology, Government of India*

TEACHING EXPERIENCE

- August – December 2023: Teaching Assistant for Introductory Electricity & Magnetism.
- August – December 2021: Teaching Assistant for Introductory Electricity & Magnetism.
- April 2025, 2026: Guest lecture on tidal disruption events in an undergraduate class on Stellar and Interstellar Astrophysics.
- April 2025: Guest lecture on tidal disruption events in a combined undergraduate and graduate class on Fluid Dynamics.

MENTORING EXPERIENCE

- August 2025 – present: Mentoring Syracuse University undergraduate student, Valentino Indelicato on a computational project related to Tidal Disruption Events. Co-authored one paper.
- August 2023 – July 2025: Mentored Syracuse University undergraduate student, Julia Fancher (currently a graduate student in the Astrophysics program at Ohio State University), on projects involving analytical and numerical modeling of Tidal Disruption Events. Co-authored two papers.
- June 2022 – August 2026: Mentored eight high school students Sarah Kapalanga, Angela Kumah, Aluel Athian, Valentino Indelicato, Ricky Reyes, Tizera Davis, Joshua Davis and Eden Price on computational projects related to Stellar Evolution and Tidal Disruption Events as part of the Syracuse University Research in Physics (SURPh) program.

PROFESSIONAL SERVICE

- Referee for ApJ, MNRAS, Science Bulletein.
- Executive Secretary for NICER C7 and C8 panel reviews.

PROFESSIONAL DEVELOPMENT

- North American Einstein Toolkit School and Workshop
  Rochester Institute of Technology, NY July 2023
- Workshop on Neutron Rich Matter on Heaven and Earth
  Institute of Nuclear Theory (INT), Seattle, WA (virtual) July 2022
- Open Science Grid User School
  University of Wisconsin, MI, Computer Sciences Department July 2022